1 Introduction

TBD

2 Equations of Motion

2.1 Analytical Derivation

Consider a ball of radius R and mass M rolling without slipping on a plane

$$H_\alpha := \{(x, y, z) \in \mathbb{R}^3 \mid F := z - \alpha y = 0\} \subseteq \mathbb{R}^3 \tag{1}$$

of slope α and unit normal field $\boldsymbol{n} := \frac{\operatorname{grad} F}{\|\operatorname{grad} F\|}$. Assume that the sphere carries a dipole moment $\boldsymbol{\mu}_0$ interacting with a homogeneous, constant magnetic field $\boldsymbol{B}$. The configuration manifold is

$$Q := H_\alpha \times \mathrm{SO}(3) \subseteq \mathbb{R}^3 \times \mathrm{SO}(3),$$

where SO(3) is the three-dimensional special orthogonal Lie group. The velocity phase space is thus given by

$$TQ = TH_\alpha \times T\,\mathrm{SO}(3) \cong TH_\alpha \times \mathfrak{so}(3) \cong \mathbb{R}^2 \times \mathbb{R}^3 \subseteq \mathbb{R}^6,$$

where T denotes the tangent bundle and $\mathfrak{so}(3)$ is the Lie algebra of SO(3). We choose the trivial global chart on $\mathbb{R}^3$ and some local smooth chart $\phi : U \subseteq \mathrm{SO}(3) \to \mathbb{R}^3$ on SO(3).[1] A motion is a function

$$(\boldsymbol{r}(t), P(t)) : \mathbb{R} \supseteq I \to \mathbb{R}^3 \times \mathrm{SO}(3),$$

where $\boldsymbol{r}(t)$ is a path in $\mathbb{R}^3$ and $P(t)$ is a path in SO(3), i.e., a time-parametrized special orthogonal matrix. Define $\boldsymbol{\theta} := \phi \circ P : I \to \mathbb{R}^3$. Our goal is to find equations of motion and their solutions

$$(\boldsymbol{r}(t), \boldsymbol{\theta}(t)) = (x(t), y(t), z(t), \theta_x(t), \theta_y(t), \theta_z(t)) : I \to \mathbb{R}^6$$

in local coordinates. Using the canonical Euclidean basis $\mathcal{B} := (\boldsymbol{e}_x, \boldsymbol{e}_y, \boldsymbol{e}_z)$, we define a non-moving (inertial) coordinate frame by choosing an arbitrary origin. Furthermore, we consider a rotating coodinate frame defined by $\hat{\mathcal{B}}(t) :=$

[1] Constraining the motion to H_α reduces the system by one dimension, but we choose not to eliminate the z coordinate since we need the solution in the full frame for numerical analysis.

$P(t)\mathcal{B}$. The angular velocity in the inertial frame $\boldsymbol{\Omega}(t) = (\Omega_x, \Omega_y, \Omega_z)^\top$ is defined in terms of the cross product matrix

$$\Omega^\times(t) = \dot{P}(t)P^\top(t) = \begin{pmatrix} 0 & -\Omega_z & \Omega_y \\ \Omega_z & 0 & -\Omega_x \\ -\Omega_y & \Omega_x & 0 \end{pmatrix}.$$

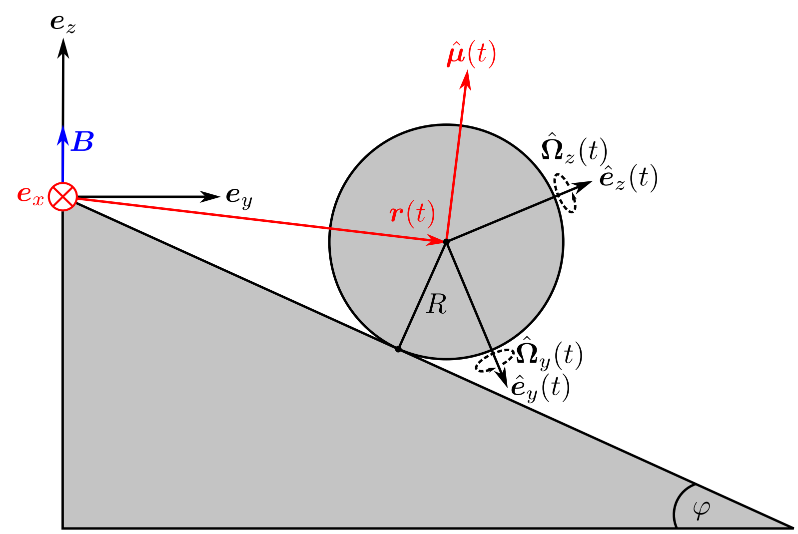

Figure 1: Sketch of the system in the yz-plane. The hat vectors are vectors of the basis $\hat{\mathcal{B}}$ while vectors without hat refer to $\mathcal{B}$. Red vectors are projected and may have non-zero $\boldsymbol{e}_x$-components. The plane H_α has an angle of inclination φ defined by $\alpha = \tan\varphi$. The outer magnetic field is kept fixed parallel to $\boldsymbol{e}_z$.

The kinetic energy form is given by

$$T(\boldsymbol{r}, \boldsymbol{\Omega}) = \frac{M}{2}\|\dot{\boldsymbol{r}}\|^2 + \frac{I_0}{2}\|\boldsymbol{\Omega}\|^2 \tag{2}$$

and the potential by

$$U(z, \boldsymbol{\Omega}) = -Mgz + \langle\hat{\boldsymbol{\mu}}(t), \boldsymbol{B}\rangle, \tag{3}$$

where $\hat{\boldsymbol{\mu}}(t) := P(t)\boldsymbol{\mu}_0$, I_0 is the inertia scalar of an ideal ball, and g is the gravitational acceleration. We obtain a Lagrangian $L(z, \boldsymbol{\Omega}, \dot{\boldsymbol{r}}) := T - U$ and follow [1] for non-holonomic systems. Assuming that there is an external force $\boldsymbol{F}_{\mathrm{e}} = \boldsymbol{F}_{\mathrm{e}}(\boldsymbol{r})$ such as friction, the corresponding Lagrange-d'Alembert principle reads[2, 3]

$$\delta \int_0^t L(z(s), \boldsymbol{\Omega}(s), \dot{\boldsymbol{r}}(s))\, ds + \int_0^t \langle \boldsymbol{F}_{\mathrm{e}}(\boldsymbol{r}(s)), \delta \boldsymbol{r}(s) \rangle\, ds = 0. \tag{4}$$

The torque can be written as

$$\boldsymbol{\tau} = \boldsymbol{\tau}_{\mathrm{em}} + \boldsymbol{\tau}_{\mathrm{e}}$$

where $\boldsymbol{\tau}_{\mathrm{em}} := \hat{\boldsymbol{\mu}}(t) \times \boldsymbol{B}$ is the magnetic torque, and $\boldsymbol{\tau}_{\mathrm{e}} := -R\boldsymbol{n} \times \boldsymbol{F}_{\mathrm{e}}$ is the torque induced by external forces. The phase space of the linear motion admits an orthogonal decomposition $\mathbb{R}^3 \cong H_{\parallel} \oplus H_{\perp}$ with respect to the normal $\boldsymbol{n}$. Thus we can decompose arbitrary $\boldsymbol{v} \in \mathbb{R}^3$ as $\boldsymbol{v} = \boldsymbol{v}_{\parallel} + v_{\perp}\boldsymbol{n}$. With the rolling constraint

$$\dot{\boldsymbol{r}} = R\boldsymbol{\Omega} \times \boldsymbol{n}, \tag{5}$$

integration by parts, and non-degeneracy of the scalar product, one obtains tangential

$$\frac{d\boldsymbol{\Omega}_{\parallel}}{dt} = \frac{1}{I_0 + MR^2}\boldsymbol{\tau}_{\parallel} - \frac{RMg}{I_0 + MR^2}(\boldsymbol{n} \times \boldsymbol{e}_z) \tag{6}$$

and normal

$$\frac{d\Omega_{\perp}}{dt} = \frac{\tau_{\perp}}{I_0} \tag{7}$$

equations for the angular acceleration from Equation 4. Taking the derivative of Equation 5, this fully determines the linear motion. Assuming Coulomb friction and aerodynamical drag by the drag equation, the external force becomes

$$\boldsymbol{F}_{\mathrm{e}} = -\nu_{\mathrm{r}} Mg \langle \boldsymbol{n}, \boldsymbol{e}_z \rangle \operatorname{sgn}(\dot{\boldsymbol{r}}) + \frac{\rho C A}{2} \|\dot{\boldsymbol{r}}\|^2, \tag{8}$$

where ν_{r} is the rolling resistance coefficient, sgn is the sign function, ρ is the fluid density of air, C is the drag coefficient, and A is the cross-sectional area of the sphere[4, 5, 6]. For numerical reasons, we reparametrize $P(t) = P((q_0(t), \boldsymbol{q}(t)))$, where

$$(q_0(t), \boldsymbol{q}(t)) : I \to \mathbb{S}^3$$

is a time-dependent unit quaternion. Subsuming, the full eqations of motion are:

$$\begin{aligned} \frac{d^2\boldsymbol{r}}{dt^2} &= \frac{5}{7M}\boldsymbol{\tau}\times\boldsymbol{n} - \frac{5g}{7}(\boldsymbol{e}_z)_\parallel, \\ \frac{d\boldsymbol{\Omega}_\parallel}{dt} &= \frac{5}{7MR^2}\boldsymbol{\tau}_\parallel - \frac{5g}{7R}\boldsymbol{n}\times\boldsymbol{e}_z, \\ \frac{d\Omega_\perp}{dt} &= \frac{5}{7MR^2}\tau_\perp, \\ \frac{d(q_0,\boldsymbol{q})}{dt} &= \frac{1}{2}(0,\boldsymbol{\Omega})\cdot(q_0,\boldsymbol{q}). \end{aligned} \tag{9}$$

2.2 Numerical Solution

The equations of motion were solved with our program `magsphere`[7]. We implemented a Runge-Kutta-Munthe-Kaas-4 method, combining the quaternion-adapted Munthe-Kaas integrator with the Runge-Kutta 4 algorithm[8, 9]. To avoid numerical instability at the discontinuities in Equation 8, we approximated

$$\mathrm{sgn}(\dot{\boldsymbol{r}}) \approx \tanh(k\|\dot{\boldsymbol{r}}\|),$$

where k is a dimensionless constant describing the steepness of tanh around the origin. For the rolling resistance coefficient, we estimated $\nu_\mathrm{r} = 0.002$ based on values of ν_r for tires on steel and rubber surfaces[10]. Assuming the International Standard Atmosphere and moderate velocities, values of $\rho = 1.2\,\mathrm{kg\,m^{-3}}$ and $C = 0.47$ were used for simulations[11, 12].

3 Results

3.1 Influence of the magnetic moment

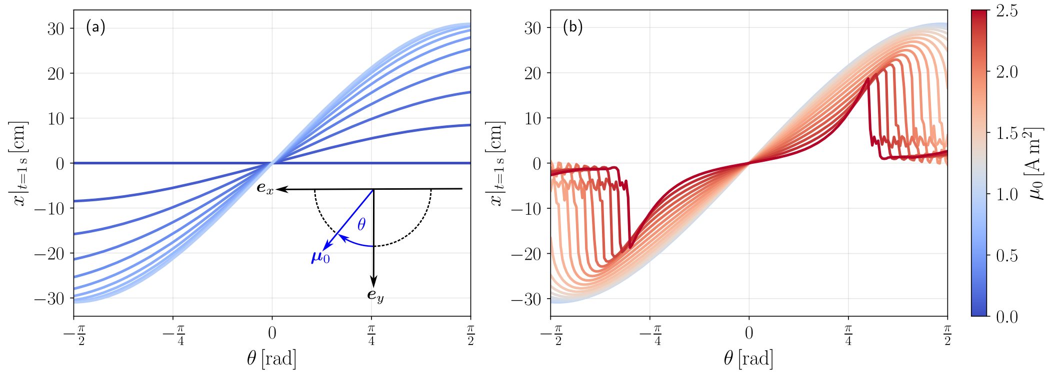

Figure 2: Final deviation $x|_{t=1\,\mathrm{s}}$ in relation to angle θ and modulus μ_0. Each graph is associated with a fixed modulus μ_0, where (a) shows curves for moduli below $1\,\mathrm{A\,m^{-2}}$ and (b) for moduli including and above $1\,\mathrm{A\,m^{-2}}$. In the small μ_0-regime (a), the final derviation increases in magnitude with θ and is antisymmetric around the origin. The high μ_0-regime (b) displays similar antisymmetry, yet different behaviour for high magnitudes of θ. With increasing modulus, critical points shift towards lower absolute values of θ, after which the curves behave erratic.

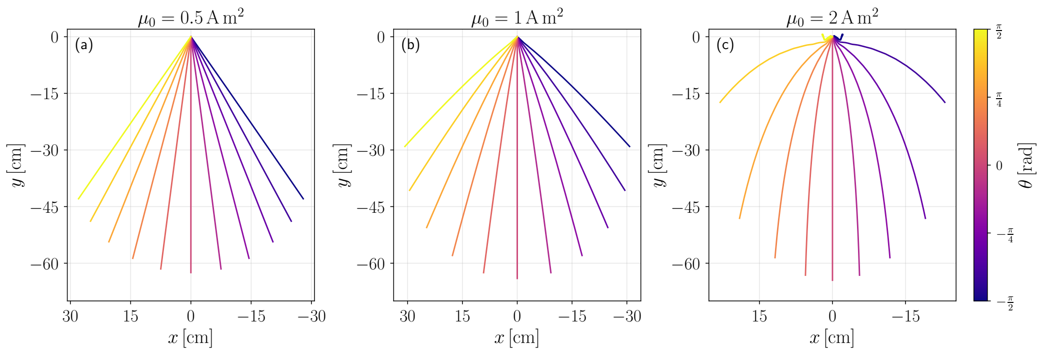

Figure 3: Trajectories on the xy-plane for selected moduli. The outward curvature of the trajectories increases with $|\theta|$ and μ_0, as anticipated. Note that in the high-modulus regime (c), the two outmost trajectories loop in on themselves as the magnetic force overcompensates the gravitational force.

We investigated the influence of the initial magnetic moment vector $\boldsymbol{\mu}_0$ on the trajectories for an incline angle of $\varphi = 10°$. Firstly focussing on the influence of the deviation of $\boldsymbol{\mu}_0$ from the plane spanned by $\boldsymbol{B}$ and the downhill direction, we parametrized

$$\boldsymbol{\mu}_0 = \mu_0(\sin\theta, \cos\theta, 0)^\top$$

with $\mu_0 := \|\boldsymbol{\mu}_0\|$ and an angle θ defined as depicted in Figure 2. We chose the value of the x-coordinate after one second as measure for the deviation of the actual trajectory from a straight downhill path. The results are also shown in Figure 2. Antisymmetric behaviour in θ is predicted by $\boldsymbol{\mu}(-\theta) = -\boldsymbol{\mu}(\theta)$. The deviation from the straight path grows non-linearly with μ_0, saturating in the low-modulus regime while reaching an extremum in the high-modulus regime. Figure 3 shows that the increasing curvature of the trajectories reduces their total length, corresponding to the saturation of x-deviation. In the high-modulus regime, the magnetic force can dominate over gravity, causing the paths to trace out loops. This explains the observed erratic behaviour beyond the global extrema in Figure 2(b).
Secondly, we parametrized

$$\boldsymbol{\mu}_0 = \mu_0(0, \sin\psi, \cos\psi)^\top$$

to investigate the influence of the xz-deviation of $\boldsymbol{\mu}_0$. The measure for the influence is the y-coordinate after one second as there is no x-deviation when $\boldsymbol{\mu}_0$ and $\boldsymbol{B}$ align in the yz-plane.

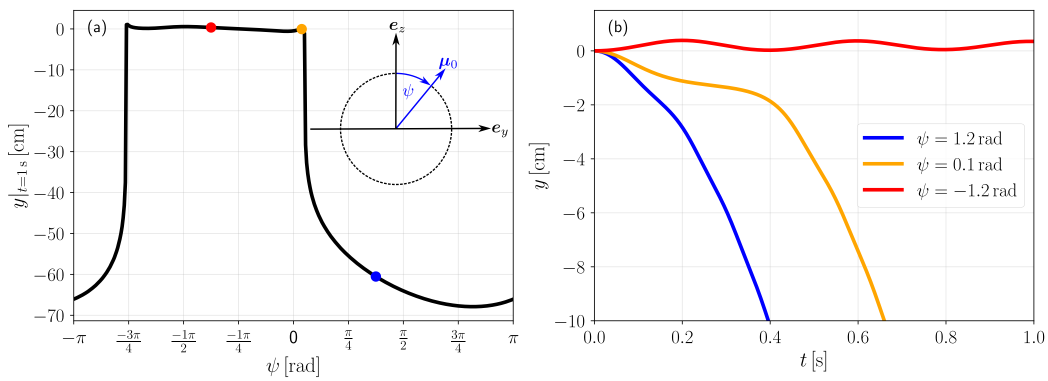

Figure 4: Final y-distance in relation to angle ψ; $y(t)$-coordinate in relation to time. Figure (a) shows three different regimes: a high-deviation regime, a low-deviation regime, and an intermediate regime. For each one of these, a representative y-trajectory is shown in (b). The low-deviation regime (blue) corresponds to a gravity-dominated downhill trajectory, the high-deviation regime (red) to a magnetic-dominated oscillatory trajectory, and the intermediate regime (orange) to a path where gravity dominates only after initial oscilltory behaviour.

Figure 4 clearly shows that for angles between about $\frac{-3\pi}{4}$ rad and 0 rad, the magnetic force fully dominates over the gravitational force, thus the ball oscillates in the yz-plane without rolling down. This behaviour transitions to oscillations which are eventually overcome by gravity at a specific point in time. The more favourable the downhill motion is for the total magnetic energy, the sooner the critial point is reached.

4 References

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